TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REGISTER

Class: **Duration time**: from 09/2026 To 03/2027

(*) Profession: Software Engineer **Specialty**: SE [x]

(*) Kinds of person make registers: Lecturer [x] Students

1. Register information for supervisor (if have)

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2. Register information for students (if have)

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3. Register content of Capstone Project

(*) 3.1. Capstone Project name:

3.1.1. English: ***Building a Smart Mushroom Cultivation Management System with Rental and Remote Monitoring Features.***

3.1.2. Vietnamese: ***Xây dựng hệ thống quản lý nuôi trồng nấm thông minh tích hợp chức năng cho thuê và giám sát từ xa***

(*) 3.2. Main Proposal Content (Including Result and Product)

a) Context:

- Environmental Sensitivity: Mushroom cultivation is highly dependent on precise micro-environmental parameters, including ambient temperature, relative humidity, and carbon dioxide (CO_2) concentrations. Fluctuations beyond optimal thresholds severely impair mycelium growth, yield, and overall crop quality.

- Limitations of Traditional Farming: Traditional mushroom farming relies heavily on manual supervision and physical inspection, which consumes significant labor resources, lacks real-time responsiveness, and introduces human error in micro-climate regulation.

- Demand for Short-Term Experience: Urban consumers demonstrate an increasing interest in organic agriculture and remote farming participation. However, they prefer short-term, experiential cultivation models (e.g., 15-day trial packages) over long-term commitments or managing complex multi-flush incubation cycles.

- Lack of Transparency: Existing agricultural services fail to provide customers with real-time sensor analytics, visual verification via direct camera live streaming, or automated harvest scheduling for their leased assets.

- Critical Need: Therefore, there is a strong need for an intelligent mushroom cultivation platform that integrates IoT-driven environmental automation, cloud services, flexible tray leasing (15, 25, 60 days), on-demand pre-conditioned spawn purchase with single-flush maximum yield extraction, and an interactive customer mobile experience.

b) Proposed Solutions:

- Unified React Ecosystem Architecture: Develop a multi-platform solution using the React ecosystem, comprising a React Mobile (React Native) Application for Customers and Shippers (Delivery Staff), a ReactJS Web Management Portal for Administrators and Farm Operators, a Node.js (ExpressJS) RESTful API core, and a PostgreSQL database.

- Tray Space Lease & On-Demand Spawn Purchase:

    - Customers select the quantity of trays and flexible Tray Lease Packages (15, 25, or 60 days).

    - Physical tray slot positions are assigned by Farm Operators based on room availability, actual tray conditions, and cultivation requirements. Customers select the quantity of trays but do not choose specific physical positions.

    - Reserved tray capacity remains dedicated to the customer for the entire package duration, even when resting empty between cultivation runs.

- Pre-Conditioned Spawn & Single-Flush Maximization:

    - Utilize pre-conditioned (fully colonized) spawn bags. Growth timelines from slit activation to pinhead sprouting are dynamically configured per species (tailored to fit within 15, 25, or 60-day leases).

    - Farm Operators slit and treat spawn bags to trigger 100% nutritional yield extraction in a single harvest flush (Single-Flush), eliminating complex multi-flush resting and incubation cycles (RESTING_INCUBATION).

- IoT Infrastructure & Automation: Deploy ESP32 microcontrollers integrated with industrial-grade temperature, humidity, and CO_2 sensors alongside automated actuators (mist sprayers, ventilation fans, LED grow lights) operating with dynamic threshold control and manual override protection (MANUAL_OVERRIDE).

- Soft-Reservation Mechanism: Integrate a 5-minute temporary resource lock (reservation_expires_at) during order creation to secure physical tray slots and spawn inventory prior to payment verification.

- Remote Monitoring & Live Streaming: Integrate Imou Cloud-based video streaming via API alongside real-time sensor telemetry dashboards to provide remote visual monitoring and environmental transparency.

- Automated Safeguards & Grace Period: Implement an automated harvest safeguard (HARVEST_AUTO_ACTION_TIMEOUT) to cold-preserve mature yields if customers fail to confirm delivery preferences within configured deadlines, alongside a complimentary grace period for late-blooming crops.

  In-House Last-Mile Delivery & Harvest Fulfillment: Once crops reach harvest maturity (HARVEST_READY), the platform supports two fulfillment workflows:

- On-site Self-Harvesting: Customers visit the farm to harvest/pick up their crops, and Farm Operators update the status directly to SELF_HARVESTED.

- App-Driven Home Delivery (Shipper Workflow): For home delivery orders, Farm Operators harvest and package the yield (PACKAGED), which is then dispatched to internal Shippers via the Mobile App. Shippers manage pickup, in-transit status (IN_TRANSIT), customer contact, and capture Proof of Delivery (PoD) photos to complete the order (DELIVERED) or handle failed delivery attempts.

c) Functional Requirements:

Administrator:

- As an Administrator, I want to manage system user accounts, assign roles (Admin, Shipper,Operator, Customer), and configure security permissions to maintain access control.

- As an Administrator, I want to assign packaged delivery orders to available Shippers (or configure auto-dispatch rules) and monitor delivery performance.

- As an Administrator, I want to manage cultivation rooms, tray inventories, and physical layout configurations across farm facilities.

- As an Administrator, I want to configure Tray Lease Duration Packages (15, 25, 60 days) and Mushroom Species Spawn Pricing/Growth Parameters.

- As an Administrator, I want to manage IoT device mappings, sensor threshold configurations, and Imou Cloud camera API integrations (Device Serial/Channel bindings).

- As an Administrator, I want to manage customer disputes, evaluate yield loss compensation, and authorize refund requests (Full/Partial).

- As an Administrator, I want to view business intelligence dashboards, system activity logs, environmental compliance reports, and export analytical data.

Farm Operator (Technician):

- As a Farm Operator, I want to receive paid spawn order notifications, verify pre-conditioned spawn bag quality, assign physical tray positions based on room availability and cultivation requirements, and update mushroom cultivation and care statuses via the Web Management Portal.

- As a Farm Operator, I want to monitor real-time temperature, humidity, and CO_2 sensor readings and receive instant environmental alert notifications.

- As a Farm Operator, I want to execute manual hardware overrides (Pumps, Fans, Lights) while automatically pausing system automation for safety.

- As a Farm Operator, I want to slit spawn bags for single-flush activation, log growth milestones, and attach verification photos.

- As a Farm Operator, I want to inspect crop maturity, mark statuses as HARVEST_READY, and trigger customer notification workflows.

- As a Farm Operator, I want to harvest mature crops, package home-delivery orders (PACKAGED) to hand over to Shippers, or mark on-site pickups directly as SELF_HARVESTED.

Customer:

- As a Customer, I want to register an account via Email OTP verification and manage personal profile/delivery address data securely.
- As a Customer, I want to select the quantity of trays and Tray Lease Duration Packages (15, 25, 60 days), with specific physical tray positions subsequently assigned by Farm Operators upon order fulfillment.
- As a Customer, I want to hold tray/spawn reservations for 5 minutes and complete payments via integrated payment gateways (PAYOS).
- As a Customer, I want to purchase pre-conditioned spawn batches, select preferred mushroom species, and track species-specific growth timelines.
- As a Customer, I want to track active cultivation progress, monitor live sensor graphs, and view real-time camera video streams of my leased tray space.
- As a Customer, I want to receive push notifications for crop milestones (sprouting, harvest readiness) and sensor alerts.
- As a Customer, I want to be notified by farm staff when crops reach harvest maturity and confirm my preferred fulfillment method (On-site Self-Harvesting or Home Delivery) for each harvest.
- As a Customer, I want to decide whether to purchase another spawn batch or leave the tray empty until the lease package expires.
- As a Customer, I want to submit feedback, report damaged deliveries via a Dispute Ticket system within 24 hours, and view full rental session history.
- As a Customer, I want to track the real-time delivery status of my harvested mushrooms (PACKAGED, IN_TRANSIT, DELIVERED), view assigned Shipper contact details, and view Proof of Delivery (PoD) photos upon completion.

Shipper (Delivery Staff):

- As a Shipper, I want to log into the Mobile Application, manage my working availability status (Online/Offline), and receive push notifications for newly assigned mushroom delivery orders.

- As a Shipper, I want to view the list of assigned delivery tasks with farm pickup details, customer delivery addresses, contact phone numbers, and integrated map navigation routes.

- As a Shipper, I want to confirm package pickup at the farm facility, updating the delivery status from PACKAGED to IN_TRANSIT (Picked Up / Out for Delivery).

- As a Shipper, I want to contact customers directly via the app, capture Proof of Delivery (PoD) photos upon handing over the package, and update the final status to DELIVERED.

- As a Shipper, I want to log failed delivery attempts (DELIVERY_FAILED) with specific reasons and evidence photos (e.g., customer unreachable, rescheduled) to trigger return-to-farm cold storage procedures.

- As a Shipper, I want to view my completed delivery history and daily fulfillment statistics.

d) Non-Functional Requirements:

- Performance: Real-time sensor data transmission latency under 2 seconds; API response time under 500ms for standard queries under concurrent usage.

- Usability: High-usability React Mobile (React Native) UI tailored for non-technical customers and on-the-go Shippers, alongside an intuitive ReactJS web dashboard optimized for farm operations.Security: Role-Based Access Control (RBAC), JWT authentication, HTTPS/TLS data encryption, transaction idempotency, and secure payment webhook processing.

- Scalability: Modular database architecture supporting horizontal scaling for additional cultivation rooms, IoT nodes, camera streams, and tenant users while decoupling tray leases from spawn execution batches.

- Availability: High system availability ($\geq 99.5\%$) with local IoT offline queuing to preserve sensor telemetry during temporary network outages.

- Reliability: Data integrity protection ensuring idempotent payment processing, strict state machine transitions, and soft-reservation rollback routines.

e) End-to-End System Workflow

Step 1: Account Registration & Tray Lease Selection

1. **User Authentication:** Customers register or log in to the Mobile Application (React Native) via Email OTP verification.

2. **Lease Package Selection:** Customers specify the desired quantity of trays and select a lease duration package (15, 25, or 60 days).

3. **Tray Capacity Verification:** The system verifies real-time available tray capacity across cultivation rooms to ensure sufficient slots for the customer's requested quantity without requiring manual position selection by the customer.

Step 2: Spawn Purchase & Soft-Reservation

1. **Mushroom Species Selection:** Customers select their preferred mushroom species (e.g., Oyster Mushroom, King Oyster Mushroom) corresponding to their leased tray slots.

2. **Temporary Lock:** The system generates an order and triggers a 15-minute soft-reservation mechanism (reservation_expires_at = NOW() + 5 mins) to temporarily hold the requested tray capacity and spawn inventory while the customer completes payment.

Step 3: Payment Processing via Payment Gateway

1. **Transaction Execution:** Customers complete payment via the integrated **PayOS** payment gateway (Bank Transfer QR / Payment Webhook).

2. **Order Fulfillment & Rollback:**

    - Upon successful payment webhook verification, the order status updates to PAID, officially activating the tray lease and spawn batch subscription.

    - If payment is not completed within 5 minutes, the automated rollback system releases the reserved tray slots and cancels the order (Soft-reservation Rollback).

Step 4: Role Assignment & Farm Operation Setup

1. **Access Control Assignment:** Administrators assign room and tray management permissions to specific Farm Operators (Staff) via the Web Management Portal (ReactJS).

2. **Order Notification:** Assigned staff receive instant web notifications for newly paid spawn orders (PAID).

3. **Tray Assignment, Spawn Placement & Activation**: Staff inspect the quality of the pre-conditioned spawn bags (100% colonized), assign specific physical tray slot positions based on room availability and environmental conditions, place the bags, and perform bag slit activation (Slit Activation) for single-flush harvesting (Single-Flush).

4. **Status Update:** Staff update the cultivation status on the Web Portal to PLANTED.

Step 5: IoT Environmental Automation & Remote Telemetry

1. **IoT Infrastructure (ESP32):**
    - Microcontrollers collect ambient temperature, relative humidity, and CO_2 sensor telemetry, transmitting updates to the server at sub-second intervals.
    - Actuators (ultrasonic mist sprayers, ventilation fans, LED grow lights) automatically trigger based on species-specific dynamic thresholds.
    - Staff retain manual override authority (MANUAL_OVERRIDE) via the web portal to handle hardware anomalies or manual maintenance.
2. **Customer Remote Monitoring:**
    - Real-time environmental sensor telemetry visualization on the mobile app.
    - Direct Imou Cloud-based live video streaming via API focused on the customer's dedicated leased tray space.
    - Automated push notifications for growth milestones (e.g., pinhead sprouting, harvest maturity).

Step 6: Harvest Notification & Simplified Fulfillment

1. Harvest Maturity Marking: When crops reach full maturity, staff mark the tray status as HARVEST_READY
2. Customer Communication: Staff notify the customer and confirm the preferred fulfillment method (Home Delivery or On-site Self-Harvesting).
3. **Fulfillment Execution:**

- **On-site Self-Harvesting**: Customers visit the farm facility to harvest their crops in person. Upon completion, staff update the status to `SELF_HARVESTED`.
- **Home Delivery via Shipper App**:

    Packaging & Dispatch: Staff harvest and package the fresh mushrooms (`PACKAGED`), and the order is assigned to an active **Shipper**.

    Pickup & In-Transit: The Shipper accepts the delivery task on the Mobile App, picks up the package from the farm, and updates the status to `IN_TRANSIT`.

Last-Mile Delivery & PoD: The Shipper navigates to the customer's address, hands over the package, uploads a **Proof of Delivery (PoD) photo** on the Mobile App, and marks the order as `DELIVERED` (or logs `DELIVERY_FAILED` for return handling if the customer is unreachable).

4. **Cycle Reset:** The tray returns to an idle/vacant state. The customer may choose to purchase another spawn batch or leave the tray empty until the lease package expires.

f) Theory & Practical:

- Software Engineering Standards: Application of Agile/Scrum software development methodology, UML 2.0 system modeling, and domain-driven architectural design (DDD) separating Space Management from Cultivation Lifecycles.
- Project Deliverables: Software Requirements Specification (SRS), Architecture Design Document, Database Schema Specification, System Test Plan & Execution Report, User Manual, Source Code Repositories, and Deployment Packages.
- Technology Stack:
    - Backend Server: Node.js (ExpressJS)
    - Database System: PostgreSQL
    - API Standard: RESTful APIs
    - Web Administrator & Operator Portal: ReactJS
    - Customer Mobile Application: React Mobile / React Native
- Customer & Shipper Mobile Application: React Mobile / React Native
- IoT Hardware Infrastructure:
    - Microcontroller: ESP32 Microcontroller Platform
    - Sensors: Temperature, Relative Humidity, and CO_2 Gas Sensors
    - Visual Surveillance: Imou IP Cameras with Imou Cloud OpenAPI / SDK Video Stream Integration.
    - Actuators & Environmental Control: Ultrasonic Mist Sprayer System, Exhaust Ventilation Fans, LED Grow Lights, Heating Elements.

g) Products (Expected Deliverables):

1. Complete RESTful API Services for Smart Mushroom Cultivation & Tray Lease Management.
2. Web Management Application (ReactJS) for Administrators and Farm Operators.
3. Cross-platform Mobile Application (React Mobile / React Native) for **Customers and Shippers**.
4. Embedded IoT Environmental Sensing and Automated Control System.
5. Production-Ready Technical Documentation (SRS, System Architecture Document, Test Plan, Database Design, Installation & Deployment Guide).

h) Proposed Tasks:

- Task Package 1: Database Schema Design (PostgreSQL separating Tray Subscriptions and Spawn Batches), RESTful API Architecture, and Payment Gateway Integration.
- Task Package 2: Frontend Web Application Development (ReactJS) for Administrator and Farm Operator Roles.
- Task Package 3: Mobile Application Development (React Mobile / React Native) for Customer Rental, Telemetry Dashboard, Camera Streaming, Shipper Last-Mile Delivery Management (Order Pickup, Route Tracking, Proof of Delivery), and Notifications.
- Task Package 4: Embedded ESP32 Firmware Development, Sensor Integration, Actuator Control Logic, and IoT Communication Protocol Setup.
- Task Package 5: End-to-End System Integration, Single-Flush Execution Testing, Staging Deployment, System Performance & Acceptance Testing.
- Task Package 6: Complete Documentation Preparation (SRS, System Design, Testing Execution Report, User Manuals, Viva Defense Presentation).

4. Other comments (propose all relative things if have):

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HCM, date 19/08/2026

Supervisor (If have)	**On behalf of Registers**
(Sign and full name)	*(Sign and full name)*